

Due: August 14, 2026

1. Solve the following linear system by the method of elimination:

$$\begin{aligned}x + 2y &= 8 \\ 3x - 4y &= 4\end{aligned}$$

2. Let

$$A = \begin{bmatrix} 2 & -3 & 5 \\ 6 & -5 & 4 \end{bmatrix}$$

What are the values of a_{12} , a_{22} , a_{23} ?

3. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix}$$

If possible, compute the indicated linear combinations:

- a) $3 \cdot (2A)$ and $6A$
 - b) $3A + 2A$ and $5A$
4. Given A defined above in problem 3, compute A^T .
5. For the following matrices, compute $a \cdot b$:

a) $a = \begin{bmatrix} 1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$

b) $a = \begin{bmatrix} -3 & -2 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

c) $a = \begin{bmatrix} 4 & 2 & -1 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix}$

d) $a = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

6. Let $a = \begin{bmatrix} -3 & 2 & x \end{bmatrix}$ and $b = \begin{bmatrix} -3 \\ 2 \\ x \end{bmatrix}$. If $a \cdot b = 17$, find x .

7. For the following exercises, let

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 4 & 0 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 \\ 2 & 4 \\ -1 & 5 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 3 & 1 \\ 3 & -4 & 5 \\ 1 & -1 & -2 \end{bmatrix}$$
$$D = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} \quad E = \begin{bmatrix} 2 & -3 \\ 4 & 1 \end{bmatrix}$$

If possible, compute:

- a) AB
- b) BA
- c) $CB + D$
- d) $AB + DE$
- e) $BA + ED$

8. Let

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \\ 0 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & -1 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$

Compute the following entries of AB :

- (a) The $(1, 2)$ entry.
- (b) The $(2, 3)$ entry.
- (c) The $(3, 1)$ entry.